North American Journal of Fisheries Management 37:729–740, 2017
© American Fisheries Society 2017
ISSN: 0275-5947 print / 1548-8675 online
DOI: https://doi.org/10.1080/02755947.2017.1327902

ARTICLE

Evaluation of Genetic Population Structure of Smallmouth Bass in the Susquehanna River Basin, Pennsylvania

Megan K. Schall*
Pennsylvania Cooperative Fish and Wildlife Research Unit, Intercollege Graduate Degree Program in Ecology, Pennsylvania State University, 413 Forest Resources Building, University Park, Pennsylvania 16802, USA

Meredith L. Bartron
U.S. Fish and Wildlife Service, Northeast Fishery Center, 227 Washington Avenue, Lamar, Pennsylvania 16848, USA

Timothy Wertz
Pennsylvania Department of Environmental Protection, Rachel Carson State Office Building, 400 Market Street, Harrisburg, Pennsylvania 17101, USA

Jonathan M. Niles
Freshwater Research Initiative, Susquehanna University, 514 University Avenue, Selinsgrove, Pennsylvania 17870, USA

Cassidy H. Shaw and Vicki S. Blazer
U.S. Geological Survey, Fish Health Branch, Leetown Science Center, 11649 Leetown Road, Kearneysville, West Virginia 25430, USA

Tyler Wagner
U.S. Geological Survey, Pennsylvania Cooperative Fish and Wildlife Research Unit, Pennsylvania State University, 402 Forest Resources Building, University Park, Pennsylvania 16802, USA

Abstract
The Smallmouth Bass *Micropterus dolomieu* was introduced into the Susquehanna River basin, Pennsylvania, nearly 150 years ago. Since introduction, it has become an economically and ecologically important species that supports popular recreational fisheries. It is also one of the most abundant top predators in the system. Currently, there is no information on the level of genetic diversity or genetic structuring that may have occurred since introduction. An understanding of genetic diversity is important for the delineation of management units and investigation of gene flow at various management scales. The goals of this research were to investigate population genetic structure of Smallmouth Bass at sites within the Susquehanna River basin and to assess genetic differentiation relative to Smallmouth Bass at an out-of-basin site (Allegheny River, Pennsylvania) located within the species' native range. During spring 2015, fin clips (n = 1,034) were collected from adults at 11 river sites and 13 tributary sites in the Susquehanna River basin and at one site on the Allegheny River. Fin clips were genotyped at 12 polymorphic microsatellite loci. Based on our results, adults sampled throughout the Susquehanna River basin did not represent separate genetic populations. There were only subtle differences in genetic diversity among sites (mean pairwise genetic differentiation index F_{ST} = 0.012), and there was an overall lack of population differentiation (K = 3 admixed populations). The greatest genetic differentiation was observed between fish collected from the

*Corresponding author: mvk10@psu.edu
Received August 25, 2016; accepted May 4, 2017

out-of-basin site and those from the Susquehanna River basin sites. Knowledge that separate genetic populations of Smallmouth Bass do not exist in the Susquehanna River basin is valuable information for fisheries management in addition to providing baseline genetic data on an introduced sport fish population.

The Smallmouth Bass *Micropterus dolomieu* is a prized sport fish and an apex predator that inhabits a variety of warmwater lotic and lentic systems. This species is native to Midwestern river systems (e.g., the Mississippi River) and the Laurentian Great Lakes. Recreational appeal has led to extensive stocking of Smallmouth Bass both within and outside of its native range in the United States (Jackson 2002) and has resulted in popular introduced sport fisheries throughout the Chesapeake Bay watershed (e.g., Potomac and Susquehanna River basins). Smallmouth Bass were first introduced into the Susquehanna River basin, Pennsylvania, in approximately 1870 by using stock from the Potomac River in West Virginia (Milner 1874; Stillwell et al. 1895). The initial stocking event was followed by multiple supplemental introductions shortly after, along with widespread stocking across the basin by local anglers (Milner 1874; Stillwell et al. 1895). Smallmouth Bass were also not native to the Potomac River but were introduced into the region from their native range in the Ohio River basin (Wheeling, West Virginia; Smith 1907; Goldsborough and Clark 1908). After its introduction, the Smallmouth Bass became a popular sport fish in the Susquehanna River; however, declines in catch rates of adult fish occurred throughout the 2000s (Arway and Smith 2013). High frequencies of clinical signs of disease (i.e., melanistic spots and lesions) and intersex conditions (i.e., testicular oocytes; Blazer et al. 2014) were also observed during that period. Such observations have raised concerns regarding the overall health of Smallmouth Bass in the Susquehanna River and have led to questions about genetic population structure. Since their introduction, there has been no evaluation of population genetic structure and genetic diversity of Smallmouth Bass within the Susquehanna River basin. In fact, the current knowledge of Smallmouth Bass population genetics in riverine systems is limited to a few studies within the native range of the species, including the Interior Highlands (Stark and Echelle 1998) and the Laurentian Great Lakes region (Borden and Stepien 2006; Stepien et al. 2007; Borden and Krebs 2009). One study examined the population genetics of both native and introduced populations of Smallmouth Bass in Virginia (Hallerman et al. 2015). Information on the population genetic structure of Susquehanna River Smallmouth Bass could provide management-relevant information about gene flow and demographic connectivity among sites within the system.

Management strategies often include both ecological and evolutionary considerations, such as the physical space that organisms utilize ("the ecological population") and the degree to which gene flow occurs ("the genetic population"; Waples and Gaggiotti 2006). Information about population genetic structure, colonization histories, ecological interactions, and mechanisms of genetic isolation may be assessed to help define management units of interest. Stocking and translocation histories of introduced species also affect geographic patterning of genetic variation (Jeschke and Strayer 2006; Hänfling 2007). For example, gene flow between introduced and native Smallmouth Bass populations was inferred in the New and James rivers, Virginia (Hallerman et al. 2015). The strength of selection pressures, including the degree of geographical or behavioral isolation, may play a role in genetic differentiation and the separation of fish into genetically distinct populations and subsequent management units.

Within the Susquehanna River basin, both river main-stem (West Branch Susquehanna River, North Branch Susquehanna River, and the Susquehanna River proper) and tributary habitats provide suitable warmwater spawning areas. Cohabitation of juvenile Smallmouth Bass spawned in tributaries and rivers has been evaluated previously, with tributary-spawned fish moving to the river during early life and potentially remaining in this habitat for an extended period (Humston et al. 2010). How fish in tributaries and main-stem rivers interact to affect population genetic structure is not known, but an understanding of this is important because fish in tributaries and main-stem rivers are often managed under different fishing regulations. Characterizing and quantifying gene flow between tributary and main-stem systems may provide important information for delineating management units and evaluating management strategies (e.g., setting fishing regulations). Therefore, the objectives of this research were to (1) investigate whether Smallmouth Bass spawning in river and tributary habitats at nearby sites were genetically distinct (local scale); and (2) evaluate whether Smallmouth Bass collected at sites across the Susquehanna River basin were separate genetic populations (regional scale).

METHODS

Study sites and fish sampling.—Smallmouth Bass were collected during the prespawn period (April–June 2015) by using towboat and boat electrofishing from 24 sites within the Susquehanna River basin (11 river sites and 13 tributary sites) and at one out-of-basin site (the Allegheny River near Franklin [ARF], Pennsylvania; Figure 1). Most neighboring river main-stem and tributary sites lacked barriers to movement. Samples were collected in four of the six Susquehanna River subbasins, including the West Branch Susquehanna River (10 sites), Middle Susquehanna River (4 sites), Lower Susquehanna River (8 sites), and Juniata River

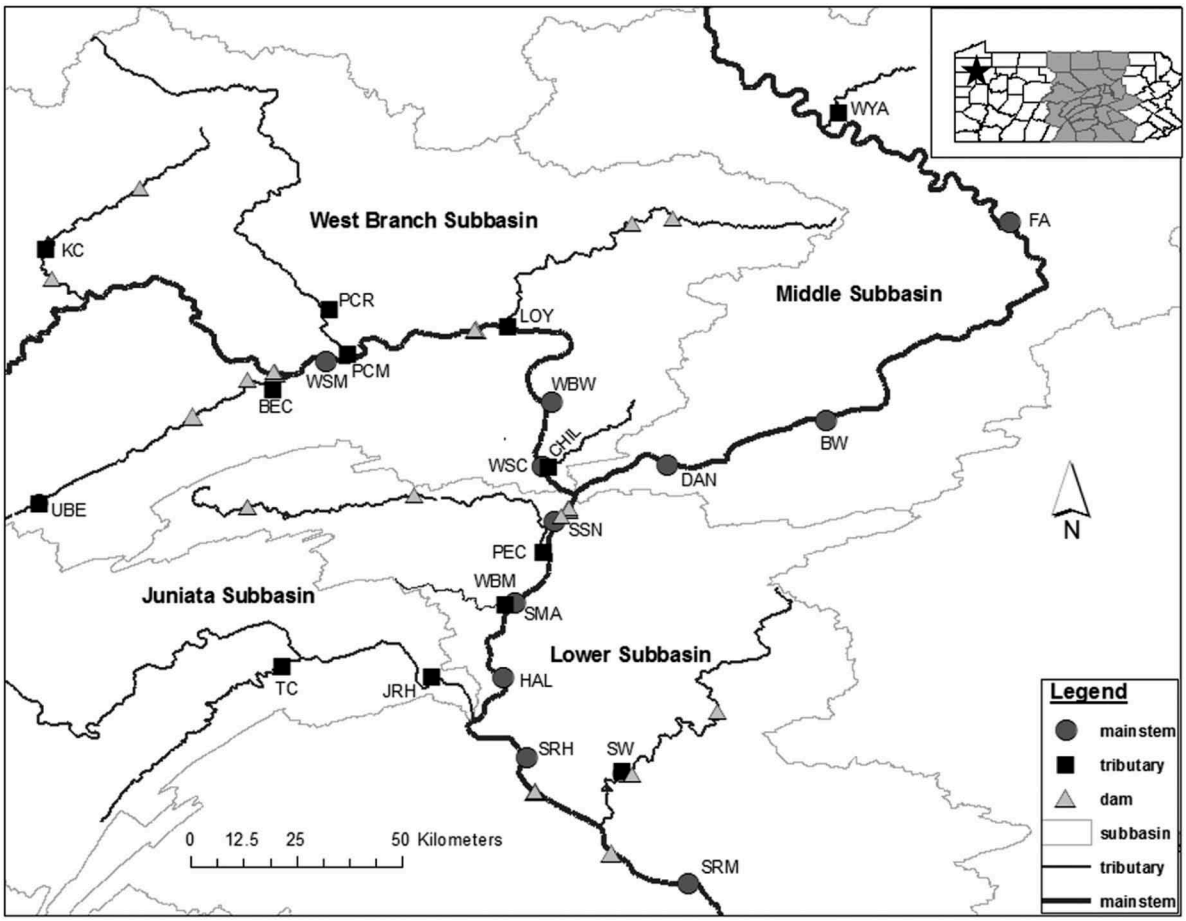

FIGURE 1. Study area in the Susquehanna River basin, Pennsylvania, where the population genetic structure of Smallmouth Bass was assessed. The out-of-basin site (Allegheny River [ARF]; black star on inset map) is in the northwestern part of the state. Samples from the West Branch Susquehanna River at Watsontown (WBW; n = 3 fish), West Branch Susquehanna River near Chillisquaque (WSC; n = 16 fish), and Kettle Creek (KC; n = 17 fish) were excluded due to low sample sizes. All other site abbreviations are defined in Table 2.

(2 sites). The two subbasins that were not sampled were the Chemung and Upper Susquehanna rivers, which are located primarily in New York State. A sample size of 50 Smallmouth Bass was targeted at each sample site, and a 9–25-mm^2 tissue sample was removed from the upper caudal fin and placed in 95% nondenatured ethanol for preservation.

Genetic analysis.—Genetic analyses were completed at the U.S. Fish and Wildlife Service (USFWS) Northeast Fishery Center in Lamar, Pennsylvania. Genomic DNA was extracted from caudal fin samples in accordance with protocols for the MagBind Tissue DNA Kit KF (Omega Biotek, Norcross, Georgia) and the KingFisher Flex Magnetic Particle Processor (Thermo-Fisher Scientific, Waltham, Massachusetts). Extracted samples were diluted 70:30 or 60:40 (reagent-grade water : eluted DNA product) depending on preliminary spectrophotometry (Gene Spec 1; Hitachi Solutions America) of DNA to reach a target of 5–30 ng/µL of diluted DNA. Ten microsatellite markers previously used to genotype Smallmouth Bass were used in this study: *Mdo1, Mdo2, Mdo3, Mdo5, Mdo7, Mdo8*, and *Mdo10* (Malloy et al. 2000); *RB7* (DeWoody et al. 1998); and *MS19* (DeWoody et al. 2000). Two additional microsatellite markers, *Mdo12* (KY056662) and *Mdo14* (KY056263) were identified from 454 GS-FLX sequencing (Roche, Branford, Connecticut) of the Smallmouth Bass transcriptome (SRX156704) using MSATCOMMANDER (Faircloth 2008). The *Mdo12* marker was derived from a 281-base-pair (bp) sequence for transcription elongation factor B, polypeptide 3, which contains a hexanucleotide repeat (AGCCTG). The marker *Mdo14* was derived from a 1,792-bp sequence for vitamin-K-dependent gamma-carboxylase, which contains a tetranucleotide repeat (AAAC). Primers for these loci were designed using PRIMER3 (Untergasser et al. 2012) and are presented in Table 1. Forward primers of microsatellite markers were fluorescently labeled with GeneScan LIZ size standard (Applied Biosystems, Thermo-Fisher Scientific) compatible dyes and multiplexed into four groups for polymerase chain reactions (Table 1). Multiplex polymerase

TABLE 1. Microsatellite markers used to investigate the genetic diversity of Smallmouth Bass in the Susquehanna River basin, Pennsylvania.

Marker	Size range (base pairs)	Number of alleles	Multiplex group	Source	Primer sequence
RB7	110–138	6	1	DeWoody et al. 1998	
MS19	104–132	13	1	DeWoody et al. 2000	
Mdo1	195–213	8	2	Malloy et al. 2000	
Mdo2	182–202	6	2	Malloy et al. 2000	
Mdo3	106–138	10	4	Malloy et al. 2000	
Mdo5	191–205	5	4	Malloy et al. 2000	
Mdo7	163–171	5	4	Malloy et al. 2000	
Mdo8	207–229	11	3	Malloy et al. 2000	
Mdo10	104–108	3	2	Malloy et al. 2000	
Mdo11	171–177	4	3	Malloy et al. 2000	
Mdo12	226–238	3	3	This study	Forward: GAGAGAGGCTATCACACGAAC Reverse: GACAGAAACAGGGACAGAGAC
Mdo14	351–355	2	2	This study	Forward: ACGGCATTTCAGATAATTTCC Reverse: CGTATCAACTTTCAAGCTAACC

chain reactions occurred in 15-µL volumes and included 1.5× GoTaq Flexi buffer, 3.75-mM $MgCl_2$, 0.3175-mM deoxynucleotide triphosphates, 0.12–0.2 µM of each forward and reverse primer, 0.08 units/µL of GoTaq Flexi DNA polymerase (Promega, Madison, Wisconsin), and 1.5 µL of diluted DNA template. Annealing temperature for polymerase chain reaction was 53°C for all markers except *RB7* and *MS19*, which had an annealing temperature of 48°C. Programs for polymerase chain reaction included denaturation at 94°C for 2 min; followed by 35 cycles of 94°C for 45 s, annealing at 53°C or 48°C for 45 s, and 72°C for 2 min; and a final step of 72°C for 5 min. The polymerase chain reaction products were diluted 1:35 (polymerase chain reaction product : distilled water), and fragment analysis was completed on an Applied Biosystems 3130XL Genetic Analyzer (Thermo-Fisher Scientific). Genotypes were scored by two separate individuals using GeneMapper version 5 (Thermo-Fisher Scientific). Quality controls were completed by selecting 96 samples (9.3% of the samples analyzed) and extracting and genotyping those samples again to compare results for consistency.

Site descriptive statistics.—Site-level analysis included testing of all markers and marker pairs for deviations from Hardy–Weinberg equilibrium and linkage disequilibrium using Fisher's exact test in Genetic Data Analysis software (Lewis and Zaykin 2001), with a Bonferroni correction for multiple comparisons (Rice 1989). Descriptive site-level statistics included the average number of alleles per locus (A), allelic richness (A_r), expected heterozygosity (H_e), observed heterozygosity (H_o), and number of unique alleles (n_u). Effective population size (N_e) was estimated using the linkage disequilibrium method in N_eEstimator version 2.01 (Do et al. 2014), with 95% confidence intervals (CIs) derived from jackknifing on loci with the lowest allele frequency set to 0.01.

Site comparisons.—A global genetic differentiation index F_{ST} was computed by using FSTAT (Goudet 2002) and the methods of Weir and Cockerham (1984), with 1,000 bootstrap replicates used to derive 95% CIs. Site comparisons employed pairwise F_{ST} (Weir and Cockerham 1984) to investigate genetic differentiation between sites using the program FSTAT, and Bonferroni correction was used for multiple comparisons of significant differences. To assess the degree of hierarchical structuring across Smallmouth Bass sites, analysis of molecular variance (AMOVA) was used to quantify genetic variability in relation to geographic distances among subbasins, among sites within subbasins, and within sites. Subbasins included four in the Susquehanna River basin (West Branch Susquehanna, Middle Susquehanna, Lower Susquehanna, and Juniata rivers) and the Allegheny River as an out-of-basin site for comparison. Sites were also grouped with respect to dam locations, and a second AMOVA was completed with only the Susquehanna River basin sites (i.e., excluding the Allegheny River site) to test for genetic variability in relation to dams (see Figure 1). Dam locations separated the study area into six regions as follows: (1) UBE; (2) BEC, WSM, PCM, and PCR; (3) LOY, CHIL, DAN, BW, FA, and WYA; (4) SSN, PEC, WBM, SMA, HAL, JRH, TC, and SRH; (5) SW; and (6) SRM (site abbreviations are defined in Table 2). We used ARLEQUIN version 3.5.2 (Excoffier and Lischer 2010) to compute a standard AMOVA with F_{ST} values and 1,000 permutations.

Site relationships and population structure.—To examine genetic relationships and genetic distances, PHYLIP version 3.695 (Felsenstein 2005) and associated programs were used to generate 1,000 replicate data sets, to calculate Cavalli-

TABLE 2. Descriptive and genetic characteristics of Susquehanna River basin sites where samples of Smallmouth Bass were collected (site = site name and location in Pennsylvania [NB = North Branch; WB = West Branch]; Abbrev = site abbreviation; Subbasin = Susquehanna River subbasin; n = number of individuals sampled; A = average number of alleles per locus; A_r = allelic richness at the lowest population size [n = 25]; H_e = expected heterozygosity; H_o = observed heterozygosity; N_e = effective population size [Inf = infinite]; 95% CI = 95% confidence interval for N_e [∞ = infinity]).

Site	Abbrev	Subbasin	n	A	A_r	H_e	H_o	N_e	95% CI
Allegheny River, Franklin	ARF	Out of basin	43	5.17	4.72	0.52	0.53	Inf	683.4, ∞
Bald Eagle Creek, Castanea	BEC	WB	46	3.92	3.72	0.52	0.51	114.2	58.0, 566.9
NB Susquehanna River, Berwick	BW	Middle	49	3.50	3.28	0.55	0.53	Inf	204.2, ∞
Chillisquaque Creek	CHIL	WB	46	3.50	3.28	0.54	0.56	Inf	164.2, ∞
NB Susquehanna River, Danville	DAN	Middle	50	3.25	3.10	0.51	0.48	1,221.1	82.2, ∞
NB Susquehanna River, Falls	FA	Middle	50	3.50	3.29	0.54	0.55	Inf	247.5, ∞
Susquehanna River, Halifax	HAL	Lower	50	3.50	3.30	0.53	0.54	Inf	225.7, ∞
Juniata River, Howe Township	JRH	Juniata	50	3.58	3.38	0.51	0.51	546.8	88.2, ∞
Loyalsock Creek	LOY	WB	35	3.42	3.29	0.54	0.54	1,040.0	58.6, ∞
Pine Creek Mouth	PCM	WB	37	3.33	3.26	0.51	0.49	60.9	27.1, 667.7
Pine Creek, Ramsey	PCR	WB	63	3.58	3.26	0.50	0.52	Inf	114.4, ∞
Penns Creek	PEC	Lower	48	3.42	3.21	0.53	0.51	Inf	231.6, ∞
Susquehanna River, Mahantango	SMA	Lower	36	3.50	3.34	0.53	0.52	Inf	95.5, ∞
Susquehanna River, Harrisburg	SRH	Lower	50	3.33	3.19	0.55	0.55	335.4	61.0, ∞
Susquehanna River, Marietta	SRM	Lower	50	4.00	3.52	0.55	0.58	45.0	23.9, 121.7
Susquehanna River, Shady Nook	SSN	Lower	49	3.50	3.31	0.54	0.53	Inf	135.2, ∞
Swatara Creek	SW	Lower	35	3.75	3.57	0.55	0.56	59.0	29.4, 264.0
Tuscarora Creek	TC	Juniata	42	3.50	3.30	0.54	0.57	Inf	70.8, ∞
Upper Bald Eagle Creek	UBE	WB	25	3.33	3.33	0.54	0.56	110.2	28.1, ∞
WB Mahantango Creek	WBM	Lower	50	3.33	3.14	0.51	0.51	342.1	68.8, ∞
WB Susquehanna River, McElhattan	WSM	WB	41	3.83	3.57	0.53	0.53	59.0	34.3, 141.3
Wyalusing Creek	WYA1	Middle	45	3.33	3.21	0.52	0.51	152.9	49.3, ∞

Sforza and Edwards (1967) chord distances, and to draw an unrooted neighbor-joining tree. TreeView (Page 1996) was used to create the final neighbor-joining dendrogram. A Mantel test and corresponding correlogram were used to evaluate the relationship between genetic distance and geographic distance. The package "ecodist" (Goslee and Urban 2007) was used to compute the Mantel test in R software (R Core Team 2013), with 10,000 permutations and 1,000 bootstrap replicates. Genetic distance was measured as $F_{ST}/(1 - F_{ST})$, and geographic distance (river kilometers [rkm] between sites) was $\log_e$ transformed as recommended by Rousset (1997). In the correlogram, site comparisons were used to determine the number of distance classes. The out-of-basin site (ARF) was omitted from this analysis because network distance was not measurable.

Reassignment of individual multilocus genotypes to original populations was tested using GeneClass2 (Piry et al. 2004) with the Bayesian reassignment algorithm (Rannala and Mountain 1997). Reassignment tests assign each individual to a predefined population based on allele frequency. Genetically unique populations show a higher proportion of correct assignments than genetically homogeneous populations. In contrast, Bayesian clustering methods group individuals based on minimization of (1) deviations from Hardy–Weinberg equilibrium and (2) linkage disequilibrium. The Bayesian clustering software STRUCTURE version 2.3 (Pritchard et al. 2000) was used to identify the most likely number of genetic clusters (i.e., populations; K) found among sample sites (K = 1–25) with an admixture model. In total, 20 runs were completed at each K-value, with 50,000 iterations used for burn-in and 50,000 iterations used for analysis. Bayesian clustering results were extracted with STRUCTURE HARVESTER (Earl and vonHoldt 2012), and the likelihood of K ($L[K]$) and ΔK (Evanno et al. 2005) were evaluated to determine the most likely number of populations. The program CLUMPP (Jakobsson and Rosenberg 2007) was used to combine separate STRUCTURE runs via the GREEDY algorithm with 1,000 permutations, and final plots were created and visualized with DISTRUCT (Rosenberg 2004).

RESULTS

Site-Level Descriptive Statistics

Eight of the sampled Smallmouth Bass (from a total of 1,034) were omitted because of missing tissue samples, poor DNA quality, or allele sizes that were outside the range expected based on the published literature. Three sites (Figure 1) were removed from all analyses due to low sample sizes, including the

West Branch Susquehanna River at Watsontown (WBW; n = 3 fish), West Branch Susquehanna River near Chillisquaque (WSC; n = 16 fish), and Kettle Creek (KC; n = 17 fish). At the remaining 22 sites, the average number of individuals collected was 45 (range = 25–63; Table 2).

In tests of Hardy–Weinberg equilibrium and linkage disequilibrium, no markers significantly deviated from equilibrium (P > 0.0002), but two marker pairs (*Mdo12* and *Mdo5; Mdo11* and *Mdo2*) significantly (P < 0.0002) deviated from expectations at one sample site per locus pair. The average A and A_r were greatest for the out-of-basin site (ARF; Table 2). In the Susquehanna River basin, A ranged from 3.25 (DAN) to 4.0 (SRM). Allelic richness adjusted for low sample size (n = 25) ranged from 3.10 (DAN) to 4.72 (ARF; Table 2). The ARF site also had the largest n_u (n_u = 8), and four sites in the Susquehanna River basin (BEC, BW, JRH, and SRM) had 1 n_u each. Estimates of H_o were between 0.50 and 0.55 (Table 2); deviations between H_o and H_e were greatest at SRM, DAN, TC, and UBE (most [77%] deviations were less than 0.02).

Finite N_e values were estimated for 12 of the 22 sites; seven sites had N_e estimates that could not be bound at the upper limit (upper 95% confidence limit was equal to infinity; Table 2). The lowest N_e estimates with fully bounded CIs occurred at SRM, WSM, SW, and LOY, with mean estimates between 45.0 (95% CI = 23.9–121.7) and 60.9 (95% CI = 27.1–667.7) individuals. The greatest N_e estimates of over 1,000 individuals occurred at DAN and LOY. The inability to estimate and determine upper confidence limits for N_e at several sites may be a result of too few alleles observed for the loci used in this analysis. This inability to estimate an upper limit on N_e has been documented elsewhere when population sizes were large (Waples and Do 2010).

Site Comparisons

The mean pairwise F_{ST} across sample sites was 0.016 (95% CI = 0.012–0.021). Pairwise comparisons of F_{ST} resulted in significant differences (P < 0.0002) between several sites (35% of all pairwise site comparisons), including the out-of-basin site ARF, which was significantly different from all Susquehanna River basin sites (mean F_{ST} = 0.072; Table 3). Overall, pairwise F_{ST} estimates within the Susquehanna River basin indicated genetic similarity among most of the sites (mean pairwise F_{ST} across Susquehanna River sites = 0.012). Three sites in the West Branch Susquehanna River basin (UBE, BEC, and PCR) had the greatest number of significant pairwise differences relative to other sites, with 20 significant differences for UBE (mean pairwise F_{ST} = 0.050), 16 differences for BEC (mean pairwise F_{ST} = 0.023), and 15 differences for PCR (mean pairwise F_{ST} = 0.022). Collectively, these three sites represented 48 (78.9%) of the 61 significant pairwise differences in F_{ST} between Susquehanna River basin sites.

An AMOVA was used to partition total genetic variation among subbasins, among sites within subbasins, and within sites. Results of the AMOVA indicated that 98% of the genetic variability was found within sites (df = 1,958). Only 1% of the variability was explained by among-subbasin groupings (df = 4), and 0.86% of the variability was explained by within-subbasin groupings (df = 17). When sites were organized based on separation by dams, 98.7% of the genetic variability was found within sites (df = 1,873). Only a small percentage of variability was explained by separation by dams (1%; df = 5) or within-dam-reach groupings (0.3%; df = 15), suggesting that dams had little influence on the overall partitioning of genetic variation among sample sites.

Site Relationships and Population Structure

The unrooted neighbor-joining tree further supported the observation of genetic similarity among many sites in the Susquehanna River basin, with only six nodes occurring in over 50% of the permutations (Figure 2). Only two sites, ARF and UBE, were separated from other sites at greater than 0.01 substitutions per base unit. Some groupings appeared to be associated with geographic distances between the respective sites, including the branching of the two Pine Creek sites (PCM and PCR) and the separation of the West Branch Susquehanna River subbasin sites (BEC and WSM) from several of the other Susquehanna River sites. However, evaluation of genetic distance and geographic distance resulted in weak relationships. The Mantel test and correlogram resulted in significant correlation scores (α = 0.05) for three of the six log-transformed distance classes. The magnitudes of those corresponding scores were 0.26 (distance class 1: mean $\log_e$[rkm] = 1.9), −0.20 (distance class 5: mean $\log_e$[rkm] = 5.3), and −0.34 (distance class 6: mean $\log_e$[rkm] = 5.7). Mantel correlation scores ranged from −1 to 1; although these scores were statistically significant, they were on the lower end of the range of possible scores, thus demonstrating that distance between sites was not necessarily an indication of the degree of genetic differentiation.

Although some statistically significant genetic differences (pairwise F_{ST}) were present among sites, it is uncertain whether the differences correspond to genetically separate populations. GeneClass2 (Piry et al. 2004) assignment tests, in which individuals were assigned to empirically derived clusters based on multilocus genotypes, resulted in only one site (ARF) with a large percentage (74%) of individuals that were reassigned correctly to the site of origin. Within the Susquehanna River basin, UBE was the only site to have greater than 50% reassignment. The remaining Susquehanna River basin sites had less than 35% reassignment (mean = 11%; range = 0–34%). Results from Bayesian clustering analysis using STRUCTURE (Pritchard et al. 2000) identified three genetically distinct population clusters (i.e., K = 3) based on both $L(K)$ and the ΔK value (Figure 3). Although individuals at the respective sites had different proportions of population clusters found within them, none of the sites

TABLE 3. Pairwise comparisons of the genetic differentiation index F_{ST} between all Smallmouth Bass sample sites within the Susquehanna River basin, Pennsylvania, and one out-of-basin site (ARF). Site abbreviations are defined in Table 2; site locations are shown in Figure 1. Bold italic values represent significant site comparisons from pairwise testing after Bonferroni correction (α = 0.0002).

Site	ARF	BEC	BW	CHIL	DAN	FA	HAL	JRH	LOY	PCM	PCR	PEC	SMA	SRH	SRM	SSN	SW	TC	UBE	WBM	WSM	WYA
ARF	0																					
BEC	***0.047***	0																				
BW	***0.077***	***0.023***	0																			
CHIL	***0.076***	***0.025***	0.000	0																		
DAN	***0.077***	***0.020***	0.002	0.004	0																	
FA	***0.071***	***0.018***	0.001	0.005	0.001	0																
HAL	***0.064***	***0.009***	0.000	0.002	0.000	0.002	0															
JRH	***0.077***	***0.023***	0.007	0.011	0.009	0.008	0.009	0														
LOY	***0.056***	*0.007*	0.000	0.000	0.000	0.001	0.000	0.005	0													
PCM	***0.062***	*0.000*	0.012	0.012	0.004	0.010	0.003	0.010	0.000	0												
PCR	***0.077***	***0.011***	***0.024***	***0.024***	***0.016***	***0.022***	***0.016***	***0.019***	0.003	0.000	0											
PEC	***0.064***	***0.020***	0.000	0.004	0.000	0.002	0.000	0.000	0.000	0.007	***0.019***	0										
SMA	***0.079***	0.015	0.001	0.010	0.003	0.000	0.004	0.000	0.003	0.003	0.010	0.001	0									
SRH	***0.070***	***0.016***	0.000	0.001	0.004	0.005	0.000	0.003	0.000	0.008	***0.021***	0.000	0.003	0								
SRM	***0.074***	***0.018***	0.000	0.001	0.004	0.007	0.002	0.003	0.000	0.002	***0.009***	0.000	0.004	0.000	0							
SSN	***0.071***	***0.025***	0.000	0.003	0.000	0.004	0.001	0.000	0.000	0.010	***0.020***	0.000	0.002	0.000	0.000	0						
SW	***0.070***	***0.016***	0.020	***0.026***	***0.032***	***0.026***	***0.020***	***0.020***	0.010	0.010	0.017	0.018	0.018	0.012	0.011	***0.028***	0					
TC	***0.079***	***0.026***	0.006	0.000	0.010	0.015	0.008	0.012	0.004	***0.018***	***0.028***	0.006	0.018	0.001	0.004	0.005	***0.023***	0				
UBE	***0.115***	***0.063***	***0.053***	***0.054***	***0.058***	***0.052***	***0.049***	***0.048***	***0.044***	***0.051***	***0.063***	***0.038***	***0.059***	***0.042***	***0.036***	***0.053***	***0.029***	***0.037***	0			
WBM	***0.071***	***0.014***	0.001	0.005	0.003	0.007	0.003	0.000	0.000	0.001	0.012	0.000	0.003	0.002	0.000	0.000	0.021	0.014	***0.064***	0		
WSM	***0.064***	0.001	0.006	***0.015***	0.009	0.010	0.003	0.012	0.000	0.002	***0.012***	***0.010***	0.008	0.006	0.006	0.011	0.010	0.014	***0.044***	0.007	0	
WYA	***0.069***	***0.028***	0.005	0.007	0.008	0.006	0.005	0.019	0.009	***0.027***	***0.044***	0.009	0.019	0.012	0.018	0.013	***0.038***	0.012	***0.058***	0.015	***0.016***	0

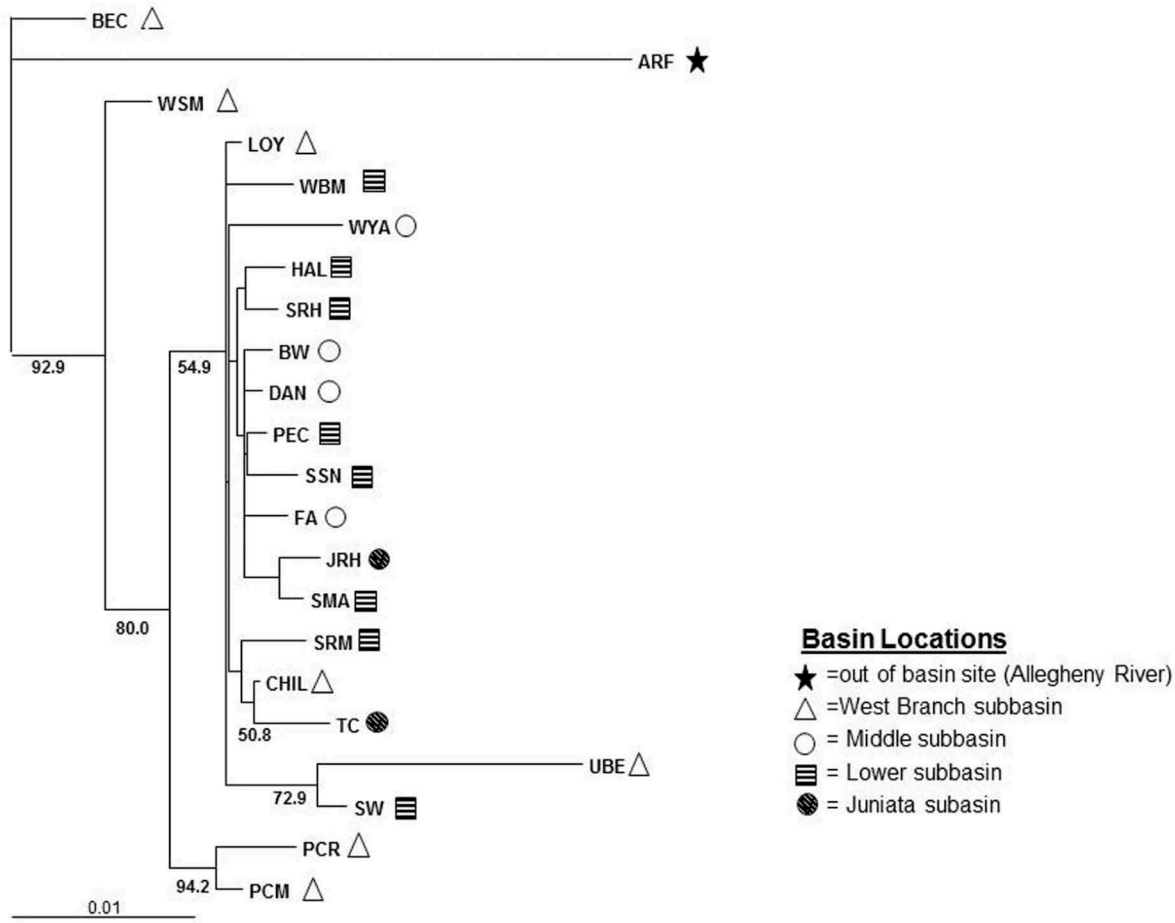

FIGURE 2. Neighbor-joining relationships based on Cavalli-Sforza and Edwards (1967) chord distances for Smallmouth Bass sampled from sites within the Susquehanna River basin (site abbreviations are defined in Table 2) and from one out-of-basin site (Allegheny River [ARF]). Symbols next to each site code correspond to basin location. Branches that occurred in over 50% of the runs include percentages associated with the branch point (scale = 0.01 substitutions per site).

appeared to have genetically distinct populations. The ARF site was the most distinct from the other sites.

DISCUSSION

We examined genetic differentiation in Smallmouth Bass among 21 sites in the Susquehanna River basin, where this species was introduced approximately 150 years ago. Overall, we found a lack of genetic differentiation among Smallmouth Bass in the Susquehanna River basin by using a combination of microsatellite markers that have successfully identified distinct populations in other studies (Stepien et al. 2007; Hallerman et al. 2015). This low genetic variability among sites was contrary to much of the previous genetic work on Smallmouth Bass population structure. Native riverine Smallmouth Bass in the Interior Highlands showed high levels of population divergence, which was consistent with historic lineages, and the fish were partitioned into separate subspecies (northern and southern forms) by using allozymes (Stark and Echelle 1998). Similarly, Stepien et al. (2007) found intermediate genetic differentiation within and among Smallmouth Bass populations sampled from the Laurentian Great Lakes region, with an overall F_{ST} estimated at 0.147. The geographic scale evaluated by Stepien et al. (2007) was much larger than the Susquehanna River, yet genetic dissimilarity (64% of F_{ST} comparisons between sites were significant after standard Bonferroni correction) was also identified within a single basin (i.e., Lake Erie). In contrast, the findings identified in the present study indicate more recent differentiation or considerable gene flow among sites in the Susquehanna River.

Smallmouth Bass populations in the Susquehanna River were founded through introductions from the same source population (Stillwell et al. 1895). These introductions do raise concern for decreased genetic diversity, although our estimates of heterozygosity (average H_o = 0.531) were similar to those of Laurentian Great Lakes populations (average H_o = 0.522; Stepien et al. 2007). Qualitative comparisons indicated lower genetic variability in the Susquehanna River basin (12 markers, 67 alleles, and 5.58 alleles/locus) than in the Laurentian Great Lakes (8 markers, 66 alleles,

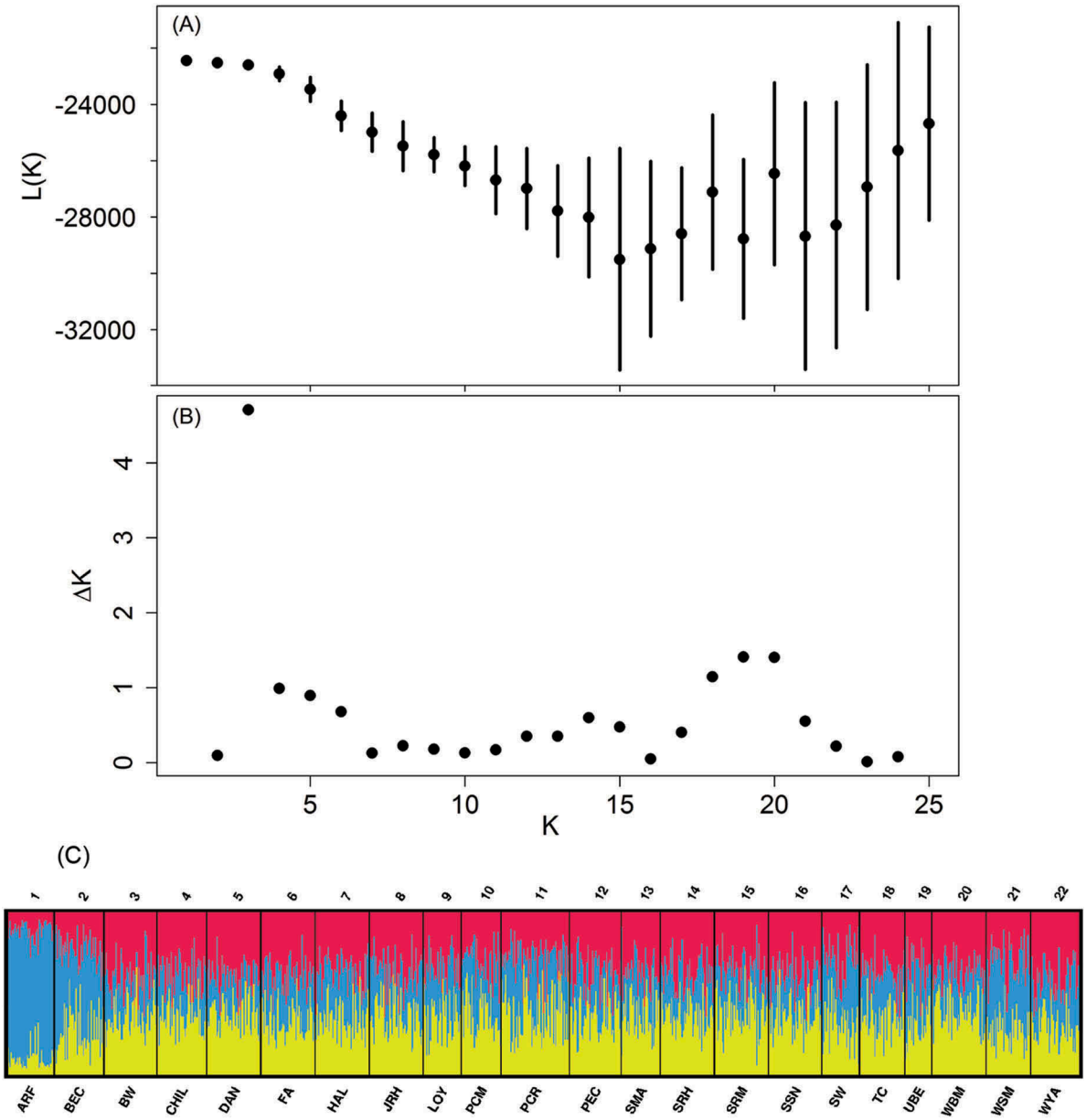

FIGURE 3. Investigation of Smallmouth Bass population structure through cluster analysis across a range of possible numbers of populations (K = 1–25) in the Susquehanna River basin and the out-of-basin site (ARF) and a graphical representation of population structure based on likely K-values (site abbreviations are defined in Table 2). Relationships between K and **(A)** the mean (±SD) likelihood of K ($L[K]$) or **(B)** ΔK (from the Evanno et al. 2005 method) were used to determine the most probable number of population clusters. **(C)** Graphical representation of three population clusters is presented (K-value of 3 was selected based on the ΔK and $L[K]$ relationships).

and 8.25 alleles/locus; Stepien et al. 2007). A similar pattern was evident when comparing the Susquehanna River sites to the out-of-basin ARF site, with heterozygosity being similar and A_r being lower. This pattern may indicate a loss of alleles over time, resulting from genetic drift. Genetic drift attributed to random loss of alleles generally affects A_r more so than heterozygosity (Allendorf et al. 2013). Loss of genetic diversity may not be an immediate concern given our estimates of genetic diversity, but introductions from the same source population may have contributed to the observed genetic similarity between sites and the limited population genetic structure. Lack of population genetic structure may also contribute to higher, unbounded estimates of N_e if populations are not genetically distinct.

Since their initial introduction into the Susquehanna River, Smallmouth Bass have not become genetically differentiated into clearly distinguishable populations. Some other stocked nonnative

fish species have diverged and formed genetically distinguishable populations in less than a century after initial introduction (e.g., Chinook Salmon *Oncorhynchus tshawytscha*; Quinn et al. 2000). The degree of genetic differentiation on shorter time scales may vary depending on the system and specific characteristics of the species of interest (i.e., the degree to which behavioral and geographical isolation occurs).

The greatest genetic differences identified for sites in this study were between the Susquehanna River sites and the out-of-basin site. This was not surprising since the Allegheny and Susquehanna rivers are not connected. Subsequent analyses, including Bayesian clustering, demonstrated the possibility for shared genetic cluster (s) with the ARF site. A historical connection is possible since Susquehanna River Smallmouth Bass were thought to have originated from the Ohio River basin (Smith 1907), the drainage into which the Allegheny River flows. Multiple genetic signatures and admixing were apparent in our cluster analysis. Black bass were historically translocated with the development of canal and railway systems (Milner 1874; Henshall 1881; Goldsborough and Clark 1908). There are reports of black bass entering into rivers through the Erie Canal system (Henshall 1881), and at some point during the 1800s, there may have been a connection between the Laurentian Great Lakes region and the Susquehanna River via the Chenango Canal (Beyer 1957). Whether Smallmouth Bass moved through this canal system into the Susquehanna River is unknown. Additionally, the known source population for the Potomac River came from the Ohio River basin near Wheeling, West Virginia, which is not proximate to the ARF site (Milner 1874; Goldsborough and Clark 1908). Even in the presence of historical connections, some of the differences between the out-of-basin site and the Susquehanna River sites may be attributable to differences in founding or initial stocks. Comparisons of genetic composition among more sites in the Laurentian Great Lakes, the Ohio River basin, and the Potomac River basin may better elucidate sources for historical stocking of the Susquehanna River and may identify the origin of the additional genetic cluster(s).

Population divergence in Smallmouth Bass could be shaped by both geographic isolation (Stepien et al. 2007) and behavioral mechanisms (e.g., spawning site fidelity; Gross et al. 1994). Geographic isolation may be attributed to natural events (e.g., glacial retreat in the Laurentian Great Lakes; Stepien et al. 2007) or physical manmade barriers (e.g., dams) that prevent gene flow. Slight genetic differentiation was apparent among a few sites in the Susquehanna River basin, including three tributary sites within the West Branch Susquehanna River subbasin. One of those three sites (UBE) was located above a flood control reservoir and could be experiencing physical isolation from the remainder of the study sites, as evidenced by the significant F_{ST} values, the highest reassignment rate, and the largest chord distance across Susquehanna River basin sites. Other types of dam (e.g., low-head dams and an inflatable dam) are also present in the Susquehanna River basin and may affect gene flow but not to the same degree as for UBE, where Smallmouth Bass are exhibiting greater genetic differentiation. For the remaining two sites (PCR and BEC), it is unclear whether physical or behavioral isolation (including spawning site fidelity) is the mechanism driving the observed genetic differences.

Variability in movement patterns of Smallmouth Bass further complicates our ability to draw parallels between systems when inferring gene flow and delineation of genetic populations. For example, there may be limited genetic structure across a large spatial extent if Smallmouth Bass are moving long distances. Movement studies on Smallmouth Bass within the species' native range have suggested varied seasonal movement in rivers, with movements ranging from less than 10 km (Todd and Rabeni 1989) to over 100 km (Langhurst and Schoenike 1990) and with homing behavior also documented in both rivers and lakes (Langhurst and Schoenike 1990; Gross et al. 1994).

Straying from natal sites and colonization of new habitat (Keefer and Caudill 2014) could also be factors in the lack of genetic structure we observed. Spawning site fidelity has been documented for Smallmouth Bass in rivers and lakes (Langhurst and Schoenike 1990; Ridgway et al. 1991; Gross et al. 1994; Barthel et al. 2008; Humston et al. 2010), but there are many variables that may influence natal homing and straying (e.g., contaminants, olfactory effects, and competition; Keefer and Caudill 2014). We chose not to estimate the migration rate using its relation to F_{ST} and N_e due to the uncertainty in the N_e estimates (i.e., a large number of unbounded estimates). Although the movement rate of Smallmouth Bass in the Susquehanna River basin has not been estimated, there have been concerns over environmental conditions and fish health within the system (Arway and Smith 2013; Blazer et al. 2014). Whether any of these environmental concerns contribute to movement or whether other mechanisms are responsible for the lack of population genetic structure is currently unknown.

Management and Future Research

Genetic management units are defined as those that exceed a predetermined value of genetic differentiation (i.e., pairwise F_{ST} level; Palsboll et al. 2006). Management decisions rely on defining management units both ecologically and genetically for angling regulations, stocking, and assessment of population status (i.e., if populations are declining; Hawkins et al. 2016). In the James River and New River watersheds, four potential genetic management units were identified for Smallmouth Bass and could be used to guide future management (Hallerman et al. 2015). In contrast, the Susquehanna River basin does not have a genetic basis for separate management units, and currently there is no information on ecologically differentiated groups of Smallmouth Bass.

A primary goal of ongoing research efforts in the Susquehanna River basin is to use genetic and ecological units to provide insights for future management and to better understand underlying mechanisms that could be influencing

fish population dynamics and health. Characterization of the population genetic structure of Smallmouth Bass in this system is one of the first steps toward doing so. Results of the present study indicate a lack of significant genetic structure in Susquehanna River basin Smallmouth Bass, which may be attributed to many factors, including introductions and movement with considerable gene flow. Genetic differentiation among a subset of sites warrants additional investigation, such as evaluation of the mechanisms for differentiation (i.e., variable selection pressure, genetic drift, and subsequent founder events) and additional out-of-basin genetic comparisons.

ACKNOWLEDGMENTS

This research was made possible through collaborative efforts among multiple research groups and partners. A research grant was received through the R. K. Mellon Freshwater Research Initiative at Susquehanna University to cover laboratory processing costs. Graduate research support was provided through the U.S. Geological Survey's (USGS) Chesapeake Bay Priority Ecosystems and Fisheries programs. Field assistance for fin clip collections was provided by personnel from the Pennsylvania Department of Environmental Protection, Pennsylvania Fish and Boat Commission, Susquehanna River Basin Commission, Susquehanna University, and USGS. We thank Shannon Julian and Jeffrey Kalie for technical assistance during laboratory work at the USFWS Northeast Fishery Center. We are grateful to Stephanie Dowell and Luke Iwanowicz for reviewing and providing advice on a previous manuscript draft. All fish handling was conducted in accordance with Protocol Number 42544 approved by the Institutional Animal Care and Use Committee at Pennsylvania State University. Use of trade names is for identification purposes only and does not imply endorsement by the U.S. Government. The findings and conclusions in this article are those of the author(s) and do not necessarily represent the views of the USFWS.

ORCID

Vicki S. Blazer http://orcid.org/0000-0001-6647-9614

REFERENCES

Allendorf, F. W., G. Luikart, and S. N. Aitken. 2013. Conservation and the genetics of populations, 2nd edition. Wiley-Blackwell, West Sussex, UK.

Arway, J. A., and G. Smith. 2013. The Susquehanna River—a fishery in decline. Fisheries 38:235–236.

Barthel, B. L., S. J. Cooke, J. H. Svec, C. D. Suski, C. M. Bunt, F. J. S. Phelan, and D. P. Phillipp. 2008. Divergent life histories among Smallmouth Bass *Micropterus dolomieu* inhabiting a connected river–lake system. Journal of Fish Biology 73:829–852.

Beyer, B. K. 1957. The Chenango Canal and the campaign for its construction. New York History 38:257–275.

Blazer, V. S., D. D. Iwanowicz, H. L. Walsh, A. J. Sperry, L. R. Iwanowicz, D. A. Alvarez, R. A. Brightbill, G. Smith, W. T. Foreman, and R. Manning. 2014. Reproductive health indicators of fishes from Pennsylvania watersheds: association with chemicals of emerging concern. Environmental Monitoring Assessment 186:6471–6491.

Borden, W. C., and R. A. Krebs. 2009. Phylogeography and postglacial dispersal of Smallmouth Bass (*Micropterus dolomieu*) into the Great Lakes. Canadian Journal of Fisheries and Aquatic Sciences 66:2142–2156.

Borden, W. C., and C. A. Stepien. 2006. Discordant population genetic structuring of Smallmouth Bass, *Micropterus dolomieu* Lacèpede, in Lake Erie based on mitochondrial DNA sequences and nuclear DNA microsatellites. Journal of Great Lakes Research 32:242–257.

Cavalli-Sforza, L. L., and A. W. F. Edwards. 1967. Phylogenetic analysis: models and estimation procedures. Evolution 21:550–570.

DeWoody, J. A., D. E. Fletcher, S. D. Wilkins, W. S. Nelson, and J. C. Avise. 1998. Molecular genetic dissection of spawning, parentage, and reproductive tactics in a population of Redbreast Sunfish, *Lepomis auritus*. Evolution 52:1802–1810.

DeWoody, J. A., D. E. Fletcher, S. D. Wilkins, W. S. Nelson, and J. C. Avise. 2000. Genetic monogamy and biparental care in an externally fertilizing fish, the Largemouth Bass (*Micropterus salmoides*). Proceedings of the Royal Society of London 267:2431–2437.

Do, C., R. S. Waples, D. Peel, G. M. Macbeth, B. J. Tillett, and J. R. Ovenden. 2014. N_eEstimator V2: re-implementation of software for the estimation of contemporary effective population size (N_e) from genetic data. Molecular Ecology Resources 14:209–214.

Earl, D. A., and B. M. vonHoldt. 2012. STRUCTURE HARVESTER: a website and program for visualizing STRUCTURE output and implementing the Evanno method. Conservation Genetics Resources 4:359–361.

Evanno, G., S. Regnaut, and J. Goudet. 2005. Detecting the number of clusters of individuals using the software STRUCTURE: a simulation study. Molecular Ecology 14:2611–2620.

Excoffier, L., and H. E. L. Lischer. 2010. Arlequin suite version 3.5: a new series of programs to perform population genetics analyses under Linux and Windows. Molecular Ecology Resources 10:564–567.

Faircloth, B. C. 2008. MSATCOMMANDER: detection of microsatellite repeat arrays and automated, locus-specific primer design. Molecular Ecology Resources 8:92–94.

Felsenstein, J. 2005. PHYLIP (Phylogeny Inference Package) version 3.6. University of Washington, Seattle. Available: http://evolution.genetics.washington.edu/phylip.html. (June 2017).

Goldsborough, E. L., and H. W. Clark. 1908. Fishes of West Virginia. Government Printing Office, Bulletin of the Bureau of Fisheries, Washington, D.C.

Goslee, S. C., and D. L. Urban. 2007. The ecodist package for dissimilarity-based analysis of ecological data. Journal of Statistical Software 22:1–19.

Goudet, J. 2002. FSTAT, a program to estimate and test gene diversities and fixation indices, version 2.9.3.2. Available: http://www.unil.ch/izea/softwares/fstat.html. (June 2017).

Gross, M. L., A. R. Kapuscinski, and A. J. Faras. 1994. Nest-specific DNA fingerprints of Smallmouth Bass in Lake Opeongo, Ontario. Transactions of the American Fisheries Society 123:449–459.

Hallerman, E. M., N. Johnson, and D. Dutton. 2015. Implications of microsatellite DNA variation upon management of Virginia Smallmouth Bass populations. Pages 615–633 *in* M. D. Tringali, J. M. Long, T. W. Birdsong, and M. S. Allen, editors. Black bass diversity: multidisciplinary science for conservation. American Fisheries Society, Symposium 82, Bethesda, Maryland.

Hänfling, B. 2007. Understanding the establishment success of non-indigenous fishes: lessons from population genetics. Journal of Fish Biology 71:115–135.

Hawkins, S. J., K. Bohn, D. W. Sims, P. Ribeiro, J. Faria, P. Presa, A. Pita, G. M. Martins, A. I. Neto, M. T. Burrows, and M. J. Genner. 2016. Fisheries stocks from an ecological perspective: disentangling ecological connectivity from genetic interchange. Fisheries Research 179:333–341.

Henshall, J. A. 1881. Book of the black bass. Robert Clarke Company, Cincinnati, Ohio.

Humston, R., B. M. Priest, W. C. Hamilton, and P. E. Bugas Jr. 2010. Dispersal between tributary and main-stem rivers by juvenile Smallmouth Bass evaluated using otolith microchemistry. Transactions of the American Fisheries Society 139:171–184.

Jackson, D. A. 2002. Ecological effects of *Micropterus* introductions: the dark side of black bass. Pages 221–232 *in* D. P. Phillipp and M. S. Ridgway, editors. Black bass: ecology, conservation, and management. American Fisheries Society, Symposium 31, Bethesda, Maryland.

Jakobsson, M., and N. A. Rosenberg. 2007. CLUMPP: a cluster matching and permutation program for dealing with label switching and multimodality in analysis of population structure. Bioinformatics 23:1801–1806.

Jeschke, J. M., and D. L. Strayer. 2006. Determinants of vertebrate invasion success in Europe and North America. Global Change Biology 12:1608–1619.

Keefer, M. L., and C. C. Caudill. 2014. Homing and straying by anadromous salmonids: a review of mechanisms and rates. Reviews in Fish Biology and Fisheries 24:333–368.

Langhurst, R. W., and D. L. Schoenike. 1990. Seasonal migration of Smallmouth Bass in the Embarrass and Wolf rivers, Wisconsin. North American Journal of Fisheries Management 10:224–227.

Lewis, P. O., and D. Zaykin. 2001. Genetic Data Analysis: computer program for the analysis of allelic data, version 1.0 (d16c). Available: http://lewis.eeb.uconn.edu/lewishome/software.html. (June 2017).

Malloy, T. P. Jr., R. A. Van Den Bussche, W. D. Coughlin, and A. A. Echelle. 2000. Isolation and characterization of microsatellite loci in Smallmouth Bass, *Micropterus dolomieu* (Teleostei: Centrarchidae), and cross-species amplification in Spotted Bass, *M. punctulatus*. Molecular Ecology 9:1919–1952.

Milner, J. W. 1874. The progress of fish-culture in the United States. Pages 524–536 *in* Report of the U.S. Commission of Fish and Fisheries 1872–1873. Government Printing Office, Washington, D.C.

Page, R. D. M. 1996. TreeView: an application to display phylogenetic trees on personal computers. Computer Applications in the Biosciences 12:357–358.

Palsboll, P. J., M. Berube, and F. W. Allendorf. 2006. Identification of management units using population genetic data. Trends in Ecology and Evolution 22:11–16.

Piry, S., A. Alapetite, J. M. Cornuet, D. Paetkau, L. Baudouin, and A. Estoup. 2004. GENECLASS2: A software for genetic assignment and first-generation migrant detection. Journal of Heredity 95:536–539.

Pritchard, J. K., M. Stephens, and P. Donnelly. 2000. Inference of population structure using multilocus genotype data. Genetics 155:945–959.

Quinn, T. P., M. J. Unwin, and M. T. Kinnison. 2000. Evolution of temporal isolation in the wild: genetic divergence in timing of migration and breeding by introduced Chinook Salmon populations. Evolution 54:1372–1385.

R Core Team. 2013. R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna. Available: http://www.R-project.org/. (June 2017).

Rannala, B., and J. L. Mountain. 1997. Detecting immigration by using multilocus genotypes. Proceedings of the National Academy of Sciences of the USA 94:9197–9201.

Rice, W. R. 1989. Analyzing tables of statistical tests. Evolution 43:223–225.

Ridgway, M. S., J. A. MacLean, and J. C. MacLeod. 1991. Nest-site fidelity in a centrarchid fish, the Smallmouth Bass (*Micropterus dolomieu*). Canadian Journal of Zoology 69:3103–3015.

Rosenberg, N. A. 2004. DISTRUCT: a program for the graphical display of population structure. Molecular Ecology Notes 4:137–138.

Rousset, F. 1997. Genetic differentiation and estimation of gene flow from *F*-statistics under isolation by distance. Genetics 145:1219–1228.

Smith, H. M. 1907. Our fish immigrants. National Geographic 18:385–400.

Stark, W. J., and A. A. Echelle. 1998. Genetic structure and systematics of Smallmouth Bass, with emphasis on interior highlands populations. Transactions of the American Fisheries Society 127:393–416.

Stepien, C. A., D. J. Murphy, and R. M. Strange. 2007. Broad- to fine-scale population genetic patterning in the Smallmouth Bass *Micropterus dolomieu* across the Laurentian Great Lakes and beyond: an interplay of behavior and geography. Molecular Ecology 16:1605–1624.

Stillwell, S. B., H. C. Demuth, L. Streuber, H. C. Ford, J. A. Dale, and D. P. Corwin. 1895. Report of the state commissioners of fisheries, year 1895. Pages 215–220 *in* Fish Commissioners of Pennsylvania, Harrisburg.

Todd, B. L., and C. F. Rabeni. 1989. Movement and habitat use by stream-dwelling Smallmouth Bass. Transactions of the American Fisheries Society 118:229–242.

Untergasser, A., I. Cutcutache, T. Koressaar, J. Ye, B. C. Faircloth, M. Remm, and S. G. Rozen. 2012. Primer3—new capabilities and interfaces. Nucleic Acids Research [online serial] 40(15):e115.

Waples, R. S., and C. Do. 2010. Linkage disequilibrium estimates of contemporary N_e using highly variable genetic markers: a largely untapped resource for applied conservation and evolution. Evolutionary Applications 3:244–262.

Waples, R. S., and O. Gaggiotti. 2006. What is a population? An empirical evaluation of some genetic methods for identifying the number of gene pools and their degree of connectivity. Molecular Ecology 15:1419–1439.

Weir, B. S., and C. C. Cockerham. 1984. Estimating *F*-statistics for the analysis of population structure. Evolution 38:1358–1370.